

COGNAC-CHATEAUBERNARD AB, France

WMO: 074120

Lat: 45.6650N Lon: 0.3158W Elev: 102 StdP: 14.64 Time Zone: 1.00 (EUC) Period: 94-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	28.4	18.6	14.1	29.7	22.0	16.5	32.2	28.0	53.0	25.0	52.8	5.5	40	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	91.0	70.1	87.5	68.4	84.1	67.1	72.4	86.3	70.6	83.4	68.9	80.3	7.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	103.1	77.3	66.3	97.2	75.4	64.8	92.2	73.7	36.0	86.9	34.5	83.7	33.0	80.6	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.8	20.0	17.6	DB	20.5	96.7	3.7	2.7	17.8	98.6	15.6	100.2	13.5	101.8	10.8	103.7
			WB	19.7	75.3	3.8	1.6	17.0	76.5	14.7	77.4	12.6	78.3	9.9	79.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	43.4	44.7	49.8	54.0	60.4	66.9	70.0	69.9	64.3	58.4	49.4	44.3
	DBStd	11.43	6.88	6.83	5.89	5.79	6.00	6.15	5.60	5.45	5.43	6.36	6.41	7.15
	HDD50	780	218	167	76	23	2	0	0	0	0	11	85	198
	HDD65	3689	670	569	471	332	169	48	16	12	77	217	467	641
	CDD50	3103	13	18	70	144	325	506	620	617	428	272	68	22
	CDD65	537	0	0	0	3	27	104	171	164	55	13	0	0
	CDH74	5126	0	0	4	59	302	1044	1600	1539	513	65	0	0
	CDH80	1776	0	0	0	6	67	363	595	604	135	6	0	0
(14) Wind	WSAvg	7.9	8.6	9.0	9.1	8.5	7.8	7.5	7.5	6.7	6.8	7.3	8.1	8.3
(15) Precipitation	PrecAvg	33.0	3.3	2.5	2.1	2.8	2.5	2.3	1.9	2.2	2.6	3.4	3.9	3.4
	PrecMax	44.1	7.9	4.9	6.5	7.6	4.8	4.4	3.7	6.5	6.7	6.7	11.1	7.6
	PrecMin	24.8	0.6	0.1	0.2	0.4	0.3	0.1	0.6	0.4	0.2	1.0	0.9	0.6
	PrecStd	4.3	1.9	1.4	1.5	1.7	1.1	1.1	0.8	1.5	1.7	1.6	2.2	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	65.2	73.4	80.2	86.3	93.9	95.2	96.3	88.8	80.1	68.1	60.3
		MCWB	53.9	53.3	56.1	61.3	65.8	71.5	70.4	71.5	67.4	65.1	59.5	55.0
	2%	DB	56.7	60.2	68.3	74.6	81.3	88.1	90.1	91.3	84.0	75.2	63.8	57.8
		MCWB	53.2	51.4	54.8	58.7	64.0	68.6	69.5	70.4	65.8	63.6	57.8	54.2
	5%	DB	54.9	56.7	64.0	70.2	77.1	84.1	86.8	86.6	79.9	71.5	61.0	56.1
		MCWB	52.0	50.4	53.3	56.7	62.5	67.3	68.4	68.5	64.6	62.5	56.7	53.3
	10%	DB	52.9	54.0	60.2	65.8	73.0	80.1	82.7	82.4	76.1	68.2	58.8	54.4
		MCWB	50.2	50.1	51.4	54.4	60.8	65.8	66.8	67.2	63.3	61.0	55.5	51.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	55.3	58.0	62.5	68.8	74.0	73.7	75.4	70.7	67.4	61.2	56.9
		MCDB	57.7	59.7	67.7	76.0	82.0	87.3	88.7	89.4	81.7	74.8	64.7	58.5
	2%	WB	54.0	53.3	56.1	60.0	66.1	71.1	71.7	72.6	68.2	65.3	59.1	55.2
		MCDB	56.0	57.0	64.8	72.5	77.9	85.1	86.5	85.9	78.3	71.9	62.0	57.1
	5%	WB	52.2	52.0	54.5	57.8	63.7	69.0	69.8	70.7	66.3	63.6	57.4	53.8
		MCDB	54.3	55.6	61.4	67.7	74.1	81.0	83.1	82.6	76.1	69.5	60.4	55.8
	10%	WB	50.6	50.5	52.9	55.6	61.6	66.8	68.2	68.7	64.7	61.9	55.6	52.1
		MCDB	52.6	53.8	58.5	63.5	70.7	77.5	79.6	79.2	73.4	67.4	58.7	54.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.3	17.0	18.3	18.3	19.4	19.8	20.5	19.8	16.5	13.1	12.0
		MCDBR	11.7	18.5	23.9	25.5	25.5	26.2	26.0	27.7	25.8	19.8	14.8	12.4
		MCWBR	8.8	11.8	13.0	12.6	11.9	11.1	10.1	10.7	11.0	10.6	9.7	9.0
	5% WB	MCDBR	10.3	14.7	18.8	22.7	22.5	23.3	23.2	23.4	21.4	16.8	13.1	10.7
		MCWBR	8.5	10.3	10.9	11.7	11.5	11.0	10.0	10.3	10.5	9.7	9.6	8.7
(40) Clear-Sky Solar Irradiance	taub		0.326	0.339	0.371	0.396	0.401	0.414	0.403	0.397	0.369	0.366	0.340	0.322
	taud		2.460	2.408	2.320	2.250	2.291	2.288	2.316	2.352	2.407	2.428	2.446	2.455
	Ebn at Noon		247	265	271	273	275	271	273	270	269	252	238	234
	Edh at Noon		22	28	35	41	41	41	39	36	32	27	22	20
(44) All-Sky Solar Radiation	RadAvg		393	697	1076	1496	1745	1957	1904	1688	1345	825	473	353
(45)	RadStd		40	76	126	146	136	143	101	115	88	70	55	45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page